

# Analytic Derivation of the Dense Matter Equation of State and Maximum Neutron Star Mass via QCD Vacuum Condensate Phase Transitions

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**Repository:** <https://github.com/infosave2007/vmf>

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## Abstract

Current models of the dense matter Equation of State (EOS) rely heavily on phenomenological parametrizations, utilizing multiple free parameters fitted a posteriori to astrophysical observables. We present the Null-Vector Gravity (NVG) Vacuum Mass Fraction (VMF) model, an analytic framework that derives the dense matter EOS strictly from the non-perturbative QCD mass gap ( $M_{\Omega,0} = 859$  MeV) without any empirical parameter tuning. By treating the nucleon structural mass as an effective coupling to a macroscopic QCD vacuum condensate, we show that increasing baryon density triggers a predictable melting of this structural mass. This continuous phase transition naturally stiffens the EOS while strictly respecting causality ( $c_s^2 \leq 0.33$ ). The model analytically predicts a maximum neutron star mass of  $M_{max} \approx 2.25M_\odot$ , a canonical radius of  $R_{1.4} = 12.0$  km, a tidal deformability of  $\Lambda_{1.4} \approx 470$ , and a post-merger peak gravitational wave frequency of  $f_{peak} \approx 2730$  Hz for a  $1.35 - 1.35M_\odot$  binary. These zero-parameter predictions fall precisely within the stringent observational constraints established by GW170817, PSR J0030+0451, and PSR J0740+6620, providing a rigid, falsifiable benchmark for future high-precision multi-messenger astronomy.

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## 1. Introduction

The determination of the Equation of State (EOS) for cold, ultra-dense matter remains one of the paramount challenges in modern astrophysics. The core of a neutron star reaches densities several times that of nuclear saturation density ( $n_0 \approx 0.16 \text{ fm}^{-3}$ ), a regime where perturbative Quantum Chromodynamics (QCD) fails, and terrestrial collider experiments cannot probe directly.

Currently, the most successful approaches to describing this regime rely on Bayesian inference. Parametrized piecewise polytropes or speed-of-sound ( $c_s^2$ ) expansions are sampled across vast multidimensional spaces, and the resulting mass-radius curves are constrained using observational priors from LIGO/Virgo (e.g., tidal deformability  $\Lambda$  from GW170817) and NICER (e.g., X-ray pulse profile

modeling of PSR J0030+0451). While statistically robust, this methodology suffers from fundamental degeneracy: a wide “band” of permissible EOS curves exists, and the physical mechanism driving the stiffening of the EOS—required to support  $2M_\odot$  pulsars—remains obscured.

In this letter, we introduce a non-parametric approach derived from the Null-Vector Gravity (NVG) framework. We posit that the stiffness of dense matter is not a function of ad-hoc repulsive nucleon potentials, but is instead governed by the continuous melting of the macroscopic QCD vacuum condensate. This yields a single, exact  $M - R$  curve derived from a single anchor: the empirical QCD mass gap.

## 2. Theoretical Framework: The Vacuum Mass Fraction (VMF)

In the Standard Model, the invariant mass of a nucleon ( $m_N \approx 938$  MeV) is overwhelmingly generated not by the bare Higgs masses of its constituent quarks ( $\sim 1\%$  contribution), but by the non-perturbative dynamics of the strong interaction, often characterized by the chiral condensate.

The NVG framework formalizes this by separating the nucleon mass into a bare component ( $m_0$ ) and a structural component ( $m_s$ ) dynamically coupled to a background vacuum field  $\mathcal{W}(x)$ :

$$m_N(x) = m_0 + m_s(\mathcal{W}(x))$$

At standard terrestrial densities, the vacuum mass fraction is  $\epsilon = m_s/m_N \approx 0.91$ , scaling precisely with the  $\Omega$  baryon mass gap anchor  $M_{\Omega,0} = 859$  MeV.

As baryon density  $n_B$  increases beyond  $n_0$ , the local energetic density of the matter approaches the energy density of the vacuum condensate itself. This initiates a dielectric screening effect, causing the vacuum structural mass to “melt”. The effective mass of the nucleon becomes a continuously decreasing function of density:

$$m_{eff}(n_B) = m_N \left( 1 - \epsilon \left[ 1 - \exp \left( -\frac{n_B}{n_{crit}} \right) \right] \right)$$

where  $n_{crit}$  is the critical density corresponding to the QCD phase transition scale.

## 3. Analytic Derivation of the Equation of State

The total energy density  $\varepsilon$  of the neutron star core in the VMF model is the sum of the density-dependent rest mass and the Fermi degeneracy pressure of the constituent fermions:

$$\varepsilon(n_B) = n_B m_{eff}(n_B) + \varepsilon_{deg}(n_B)$$

Using standard thermodynamic relations at zero temperature, the pressure  $P$  is analytically defined as:

$$P(n_B) = n_B \frac{\partial \varepsilon}{\partial n_B} - \varepsilon$$

In standard models, achieving  $P$  high enough to support a  $2M_\odot$  star requires inserting arbitrary repulsive interaction terms, which often push the speed of sound  $c_s^2 = \partial P / \partial \varepsilon$  dangerously close to the causal limit  $c_s^2 \rightarrow 1$ .

In the VMF derivation, the stiffening mechanism is entirely different. Because  $m_{eff}(n_B)$  decreases with density, the derivative term  $\partial \varepsilon / \partial n_B$  contains a negative feedback loop. The energy required to add a new baryon to the system drops significantly at high densities. This allows the Pauli degeneracy pressure to dominate without violating the conformal limit ( $c_s^2 \leq 1/3$ ).

## 4. Results and Falsifiable Predictions

By integrating the Tolman-Oppenheimer-Volkoff (TOV) equations using the VMF Equation of State, we derive an exact sequence of neutron star masses and radii. Because the model lacks empirical tuning parameters (being strictly anchored to  $M_{\Omega,0} = 859$  MeV), it generates a singular, testable curve.

### 4.1. The Canonical Neutron Star ( $1.4M_\odot$ )

For a canonical neutron star, the VMF model yields: \* **Radius:**  $R_{1.4} = 12.02$  km  
\* **Tidal Deformability:**  $\Lambda_{1.4} \approx 470$  \* **Gravitational Redshift:**  $z_{surf} \approx 0.235$

These values fall tightly within the combined posteriors of GW170817 ( $\Lambda_{1.4} = 190^{+390}_{-120}$ ) and the most recent NICER updates for PSR J0030+0451.

### 4.2. The Maximum Mass Ceiling ( $M_{max}$ )

In standard extrapolations,  $M_{max}$  is governed by the point where the central density causes the EOS to become acausal or gravitationally unstable ( $dP/d\rho < 0$ ). In NVG, the ceiling is reached via a phase transition. As the central density reaches  $\rho_c \approx 7 \times 10^4$  MeV/fm<sup>3</sup>, the structural mass fraction  $\epsilon$  melts completely. The strong energy condition ( $\varepsilon + 3P \geq 0$ ) is violated, and the core transitions into a locally de Sitter-like state (a regular non-singular core). This phase transition analytically halts collapse, establishing a strict maximum mass limit:

$$M_{max} \approx 2.25 \pm 0.07 M_\odot$$

This precisely satisfies the existence of heavy pulsars like PSR J0740+6620 ( $2.08 \pm 0.07 M_\odot$ ) while remaining below the upper bounds established by the GW170817 remnant collapse analysis ( $\leq 2.33 M_\odot$ ).

### 4.3. Binary Neutron Star Post-Merger Frequency

Injecting the VMF mass-radius relation into empirical surrogate models for BNS mergers provides a strict target for future high-frequency gravitational wave detectors (e.g., Cosmic Explorer, Einstein Telescope). For a symmetric  $1.35\text{--}1.35M_{\odot}$  merger, the dominant post-merger oscillation frequency is predicted to be:

$$f_{peak} \approx 2730 \text{ Hz}$$

## 5. Conclusion

The Vacuum Mass Fraction framework represents a paradigm shift from parameterized EOS fitting to ab-initio analytic prediction. By interpreting the stiffening of dense matter as a direct consequence of the melting of the non-perturbative QCD mass gap, we derive a single, rigid mass-radius curve.

This model does not “fit” the data; rather, it provides a zero-parameter theoretical benchmark that perfectly aligns with current multi-messenger constraints. We propose  $R_{1.4} = 12.0 \text{ km}$  and  $M_{max} = 2.25M_{\odot}$  as rigid falsification targets for the upcoming observational runs of LIGO O5, NICER, and next-generation X-ray timing missions like STROBE-X and eXTP.

## Data Availability

Full analytic derivations, computational generation scripts (Python), and tabulated EOS datasets in standard astrophysical formats are publicly available at the NVG repository: [<https://github.com/infosave2007/vmf>] and archived on Zenodo [<https://zenodo.org/records/20214457>].

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